



Ask a Question



Replication

In this lesson, you will learn about replication as a high availability mechanism.

We'll cover the following



- Replication – Active-active HA mode
- Geographical distribution of workload

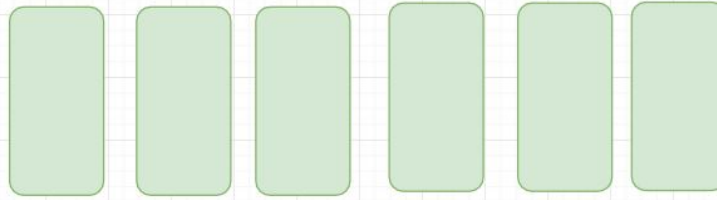
Replication – Active-active HA mode

Replication means having a number of similar nodes running the workload together. There are no standby or passive instances. When a single or a few nodes go down, the remaining nodes bear the load of the service.



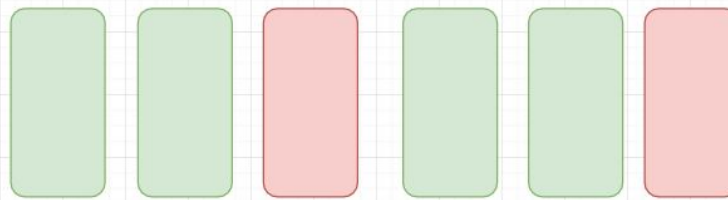


Workload - Website



Instance Fleet Sharing the Load of Running the Workload

Workload - Website



Two Nodes Go Down, the Load is Distributed Amongst the Remaining Nodes. The Service as a Whole is Still Up

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High Availability - Replication

This approach is also known as the *Active-active* high availability mode. In this approach, all the server instances are active at any point in time.

Geographical distribution of workload

As a contingency for natural disasters, regional power outages, and other big-scale failures, data center workloads are spread across different data centers across the world in different geographical locations.

This eliminates a single point of failure in the context of data center zones. If a natural disaster wipes out a few data centers in a certain zone, other data centers in different geographical zones are still powering the application. ?
Also, the latency is reduced significantly due to the proximity of data to the user due to the global replication of data centers.

